Roberta Beatrix Southwood

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PROFESSIONAL SUMMARY

Data scientist with a strong foundation in applied mathematics and computational research, bringing experience across federal and academic data analytics settings. Skilled in Python, R, and SQL, with a track record of building reproducible data workflows, developing predictive models, and communicating findings to both technical and non-technical audiences. Published researcher with two graduate-level data science research roles including a NASA internship.

SKILLS

Programming Languages:
Python (NumPy, pandas, SciPy, Matplotlib, Seaborn), R (tidyverse, ggplot2, Shiny), SQL (MySQL, PostgreSQL), MATLAB

Tools & Technologies:
Git/GitHub, Jupyter Notebook (Anaconda), Tableau, ArcGIS, LaTeX, APIs, Microsoft Office, Google Workspace

EDUCATION

American University — Washington, DC
Master of Science in Data Science. GPA: 4.0/4.0. — *December 2026*

Michigan State University — East Lansing, MI
Bachelor of Science in Mathematics, Minor in Computational Mathematics, Science, and Engineering (CMSE). GPA: 3.5/4.0. — *May 2025*

EXPERIENCE

Teaching Assistant, Statistical Programming in R / Data Science / Biostatistics — January 2026 – Present
American University — *Washington, DC*
- Assist students using R Studio, reinforcing data cleaning, script debugging, exploratory analysis, and reproducible workflows.
- Grade assignments and provide detailed feedback on statistical reasoning, code quality, and interpretation of results.
- Hold office hours to support students, translating complex statistical concepts into clear, actionable steps.

Data Science Intern — June 2026 – August 2026
National Aeronautics and Space Administration (NASA) Headquarters — *Washington, DC*
- Analyzed 10 years of demographic data on NASA Announcement of Opportunity research proposal teams, identifying trends aligned with Science Mission Directorate (SMD) leadership priorities.
- Developed visualizations and statistical analyses for internal and external research publications, including comprehensive alt text.
- Collaborated with the SMD Data Science team and presented findings to NASA Headquarters SMD leadership.
- Built reproducible data workflows and automation to streamline analysis replication and future updates.

Graduate Researcher, Data Science — August 2025 – May 2026
American University — *Washington, DC*
- Designed a data-driven research project applying sheaf modeling and topological methods to ecosystem analysis.
- Built a Python predictive model of wolf-moose predator-prey dynamics using six decades of real-world data.
- Evaluated model performance and accuracy using statistical methods.
- Presented research at StatConnect 2026 and the American University Mathias Student Research Conference.

Undergraduate Researcher, Computational Data Modeling — June 2024 – August 2024
Virginia Institute of Marine Science — *Gloucester Point, VA*
- Modeled mercury transport in Galveston Bay during Hurricane Harvey using the Regional Ocean Modeling System.
- Applied computational methods to run large-scale numerical simulations on HPC systems.
- Processed and analyzed large-scale model output using Python, MATLAB, and NetCDF.
- Presented findings at the American Geophysical Union conference; contributed to a published manuscript in JGR Oceans.

Undergraduate Learning Assistant, Calculus I — August 2023 – May 2025
Michigan State University — *East Lansing, MI*
- Led data-informed review sessions and targeted problem-solving support to improve student outcomes.
- Consistently received top evaluation scores for communication and engagement.

Undergraduate Researcher — May 2023 – July 2023
Michigan State University Topology Summer Program — *East Lansing, MI*
- Translated mathematical theory into an analytical report and delivered technical presentations to diverse audiences.

PUBLICATION

- Du, Z., Harris, CK., Southwood, R., Yin, D., Bao, D., Ye, F., Dellapenna, TM., Xu, K., Xue ZG. (2026) Redistribution of Sediment and Particulate Mercury Within Upstream Reaches of Galveston Bay in Response to Hurricane Harvey. JGR Oceans, 131(8). https://doi.org/10.1029/2025JC023581